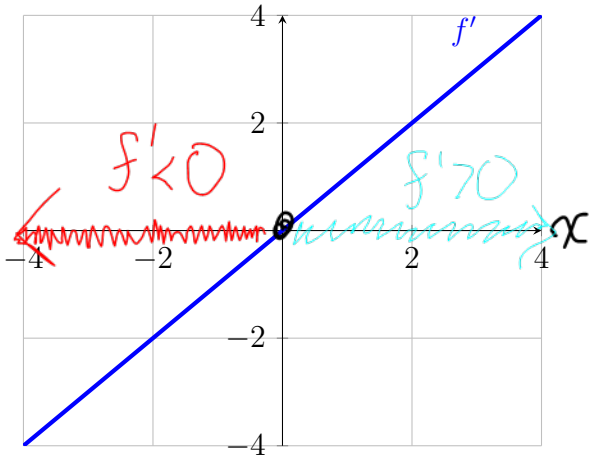
Goal: Recover information about a function f from the graph of f'.

1 Implementing the I/D and 1st Derivative Test

Problem 1. *Use the graph of f' to determine the following about f:*



(a) Intervals of increase/decrease

Crit. Nums: $x=0$



Inc: On $(0,\infty)$

Dec: On $(-\infty,0)$

(b) Positions of relative max/mins

Rel Min: $x=0$

Rel Max: None

by the 1st Derivative Test

Crit. Nums ← $f'=0$ ← when f touches the x-axis

$f'<0$ ← f' is below the x-axis

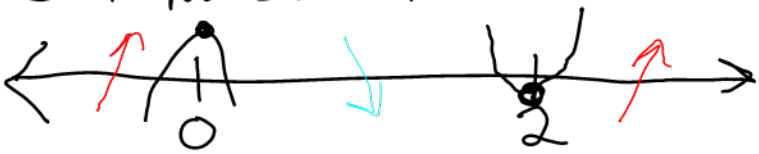
$f'>0$ ← f' is above the x-axis

Problem 2. *Use the graph of f' to determine the following about f:*



(a) Intervals of increase/decrease

Crit Nums: 0, 2



Inc: $(-\infty,0) \cup (2,\infty)$ Dec: $(0,2)$

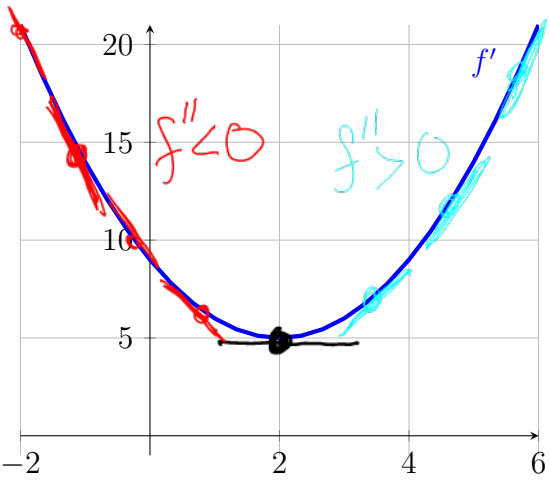
(b) Positions of relative max/mins

Rel Max at $x=0$

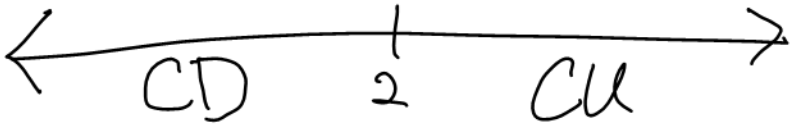
Rel min at $x=2$

2 Implementing the Concavity and 2nd Derivative Test

Problem 3. *Use the graph of f' to determine the following about f:*



(a) When f is concave up VS. concave down



CU on $(2, \infty)$

CD on $(-\infty, 2)$

(b) Position of any inflection points

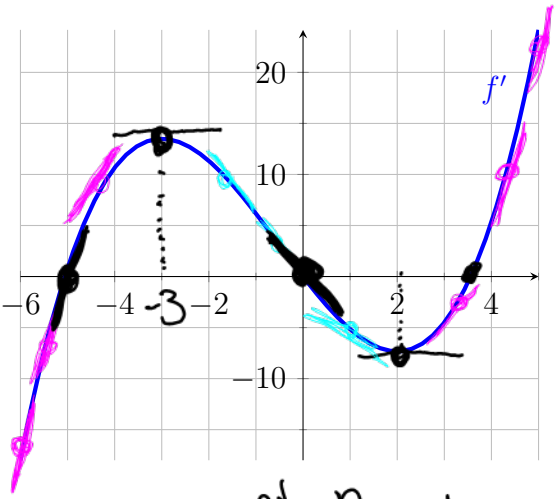
There is an IP at $x = 2$



$f'' = 0 \leftarrow [f']' = 0 \leftarrow$ When the slope of tangent lines to f' flat line

$f'' < 0 \leftarrow$ Slope of tangent lines are negative $\leftarrow$ The line "point down"

Problem 4. *Use the graph of f' to determine the following about f:*



(a) When f is concave up VS. concave down



CU on $(-\infty, -3) \cup (2, \infty)$

CD on $(-3, 2)$

(b) Position of any inflection points

IPs at $x = -3, 2$

(c) Implement the 2nd Derivative Test

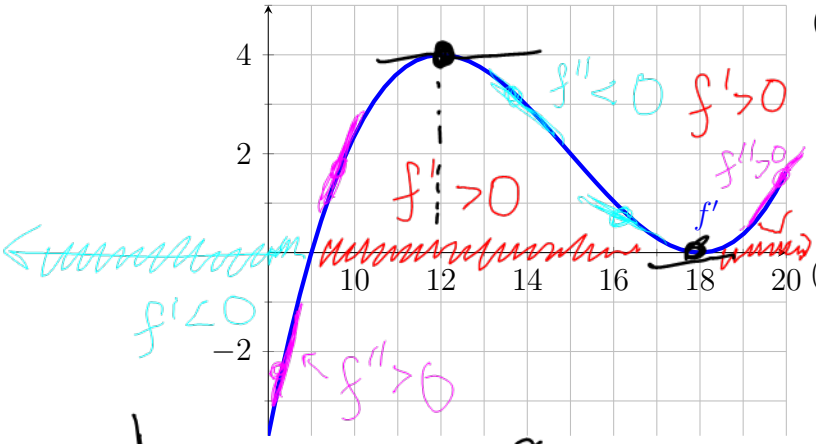
Crit Nums: $x = -5, 0, 3 + \frac{1}{2}$

$f''(-5) > 0 \cup \Rightarrow$ Rel. Min at $x = 5$

$f''(0) < 0 \cap \Rightarrow$ Rel max at $x = 0$

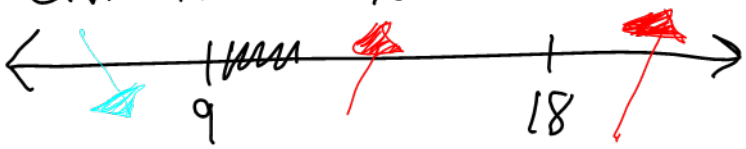
3 Putting Everything Together

Problem 5. *Use the graph of f' to determine the following about f:*



(a) Intervals of Increase/Decrease

Crit. Nums: $x = 9, 18$



(b) When f is concave up VS concave down





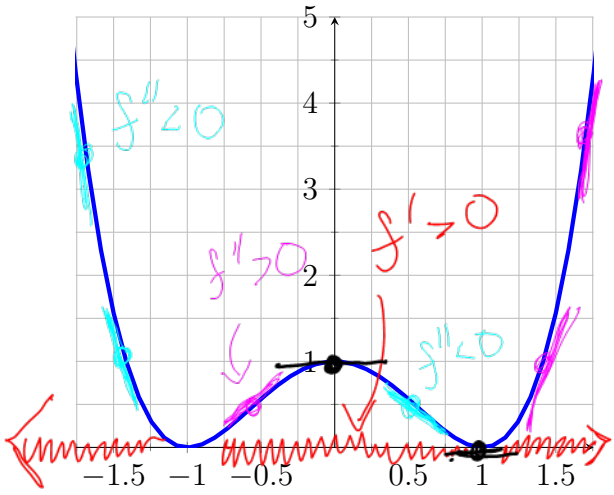
(c) State the locations of any relative max/mins as well as inflection points

Rel Max: None

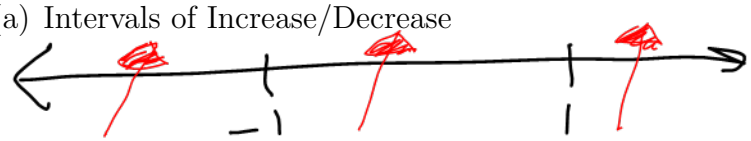
Rel Min: $x=9$

IPs: $x = 12, 18$

Problem 6. *Use the graph of f' to determine the following about f:*



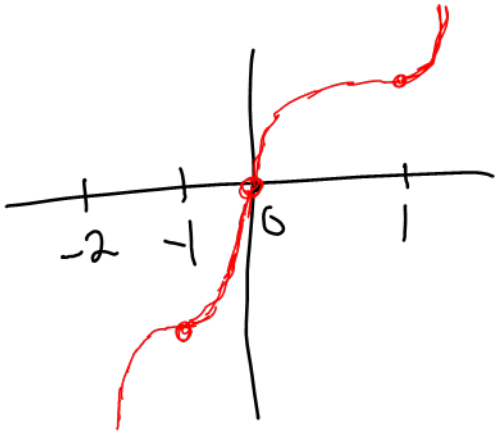
(a) Intervals of Increase/Decrease



Inc on $(-\infty, \infty)$

(b) When f is concave up VS concave down



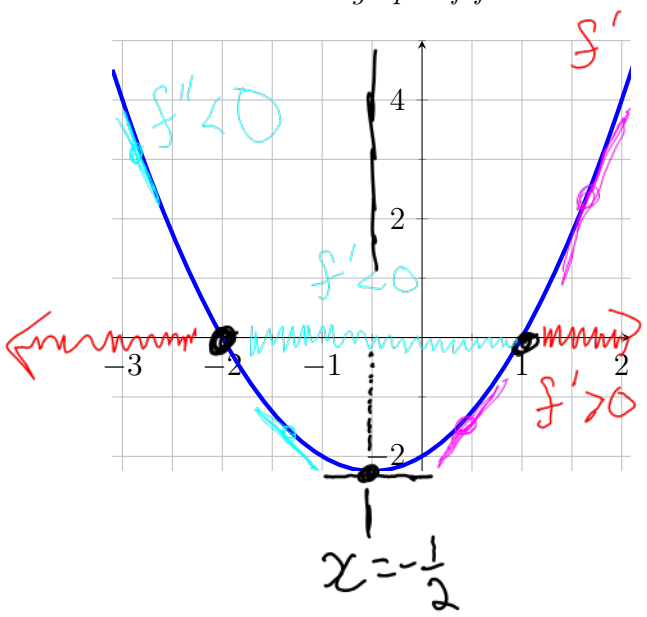


(c) State the locations of any relative max/mins as well as inflection points

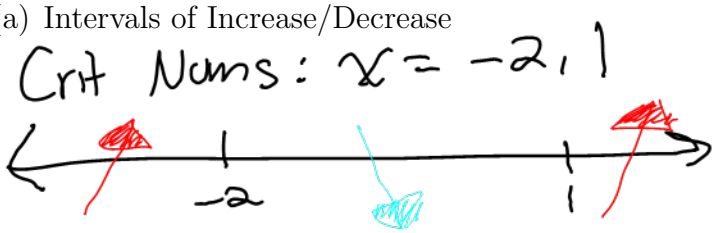
Rel Max / Mins: None

IPs: $x = -1, 0, 1$

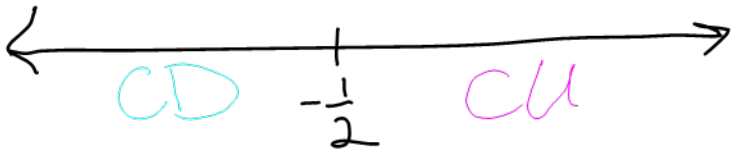
Problem 7. *Use the graph of f' to determine the following about f:*



(a) Intervals of Increase/Decrease



(b) When f is concave up VS concave down

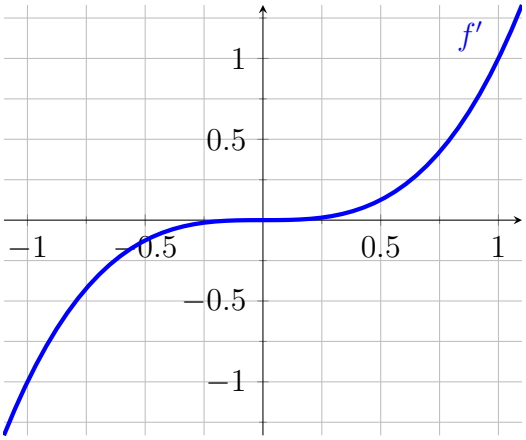


(c) State the locations of any relative max/mins as well as inflection points

Rel Max: $x=-2$ | IP at $x=-\frac{1}{2}$

Rel Min: $x=1$

Problem 8. *Use the graph of f' to determine the following about f:*

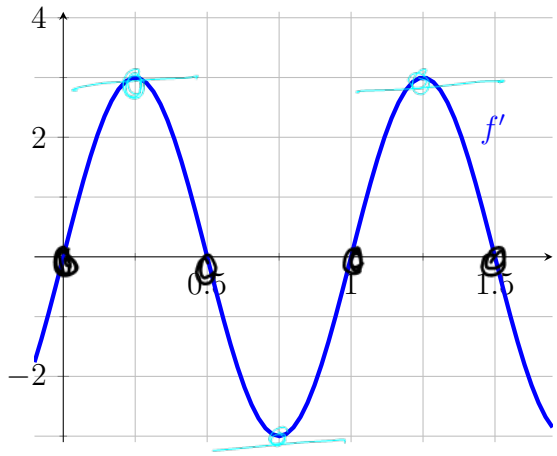


(a) Intervals of Increase/Decrease

(b) When f is concave up VS concave down

(c) State the locations of any relative max/mins as well as inflection points

Problem 9. *Use the graph of f' to determine the following about f:*



(a) Intervals of Increase/Decrease

(b) When f is concave up VS concave down

(c) State the locations of any relative max/mins as well as inflection points